

A Comparative Genomic Framework for Identifying Candidate Functional Sites in the European Bison Genome

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Can we see the effect of a recent bottleneck in the genome of the European bison? We identified evolutionarily constrained sites across the genome and found that European bison has changed more of them than its closest relatives.

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Background

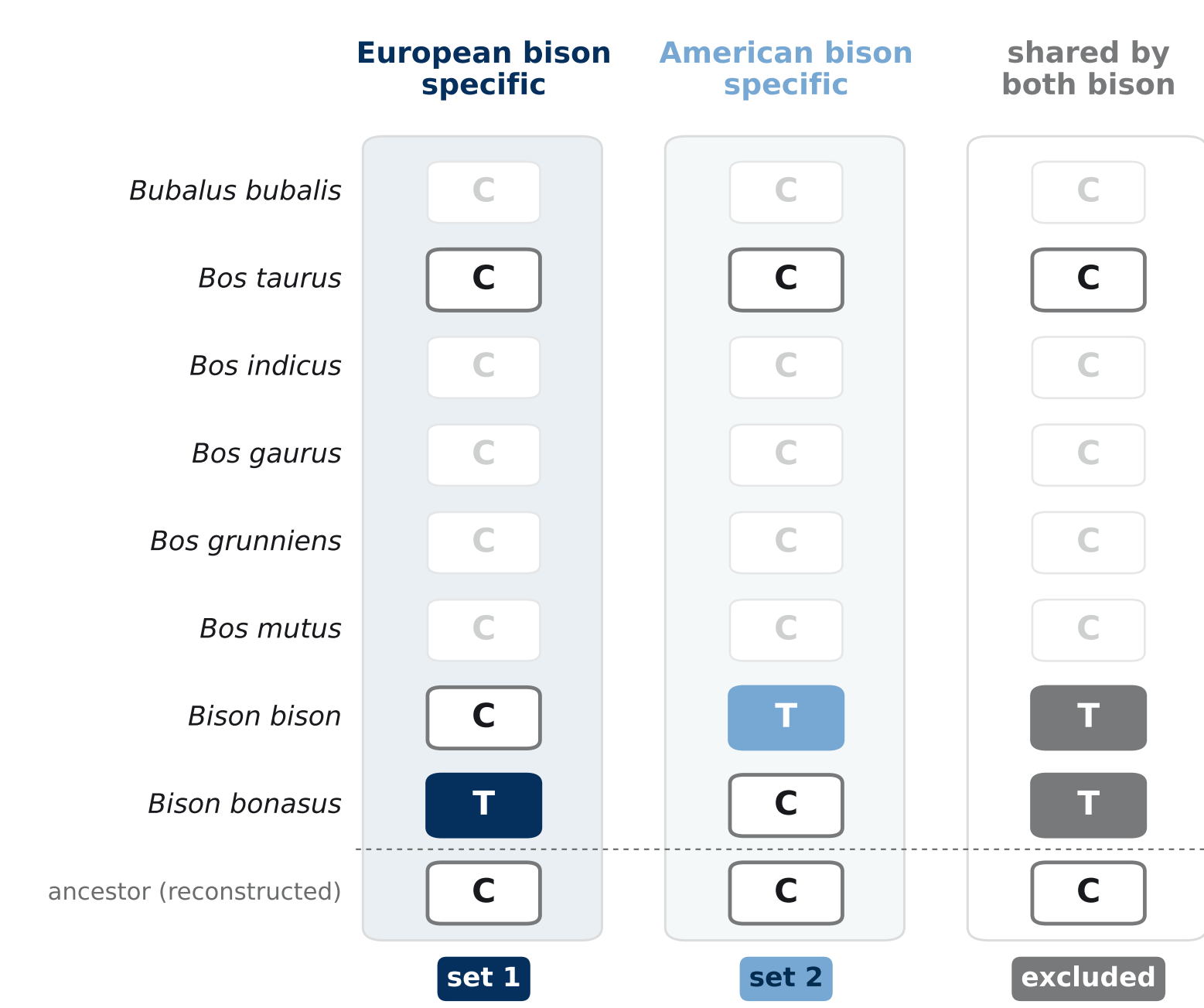
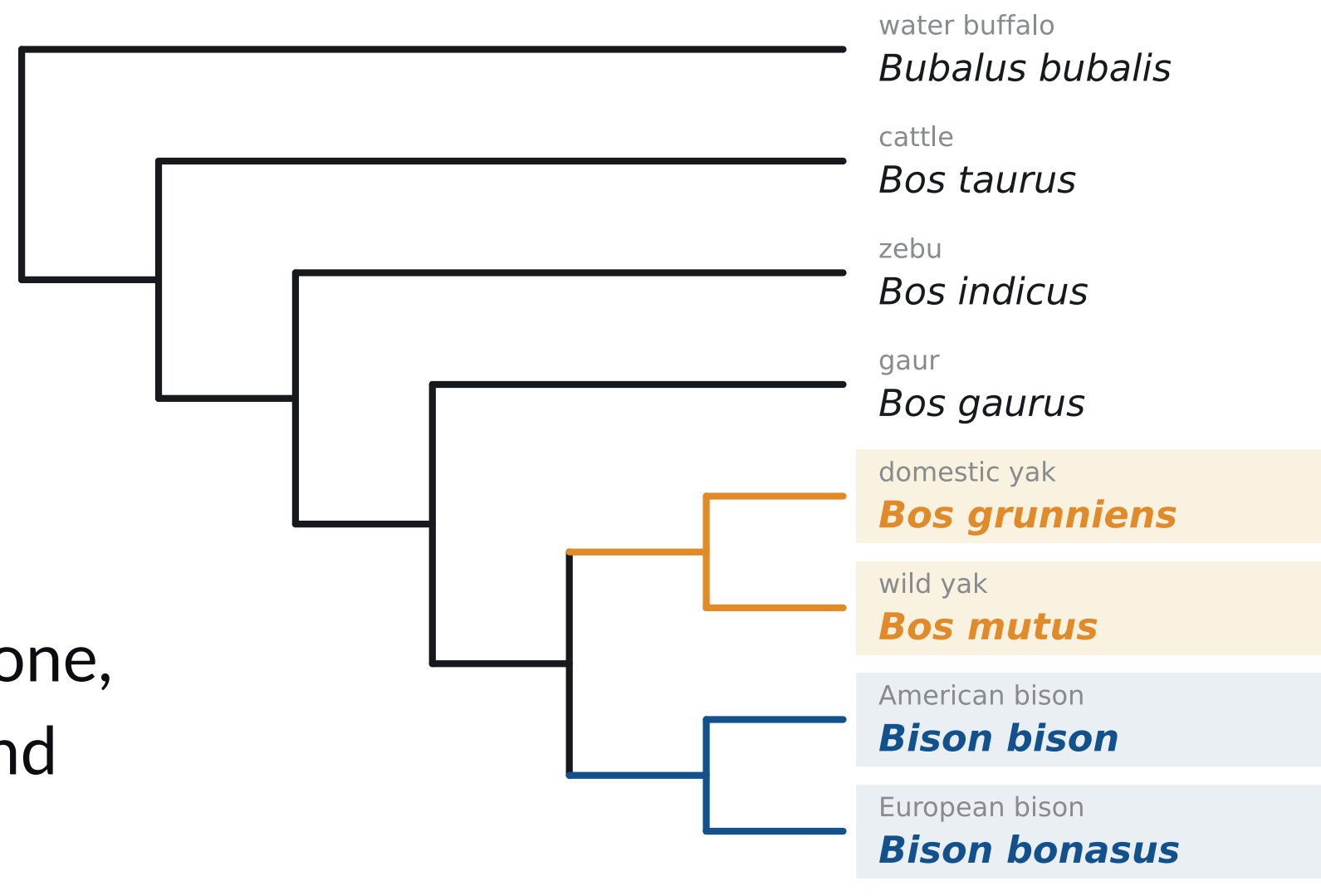
- Bison bonasus: extinct in the wild 1921, rebuilt from 12 founders
- Very low effective population size: selection loses power and harmful mutation can drift to fixation
- Non-model species: no map of which genome positions matter

Aim: find candidate functional sites from genome alignment alone, then test whether the bottleneck relaxed purifying selection and increased the rate of evolution.

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Whole genome alignment

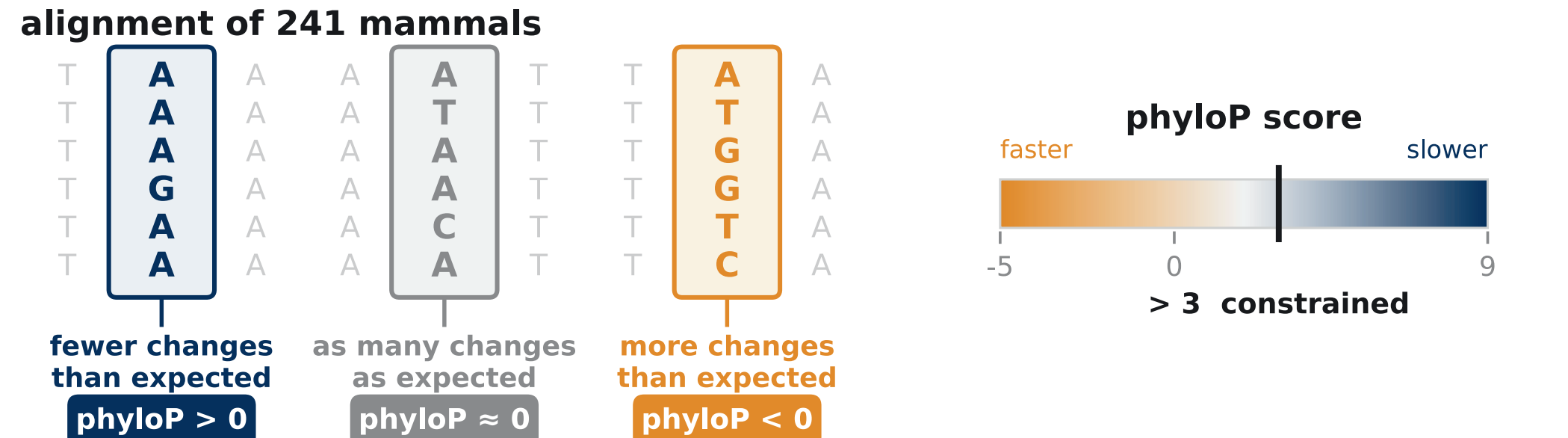
- Cactus alignment of 8 Bovini genomes with reconstructed ancestral states, used to extract species-specific changes in two contrasts.



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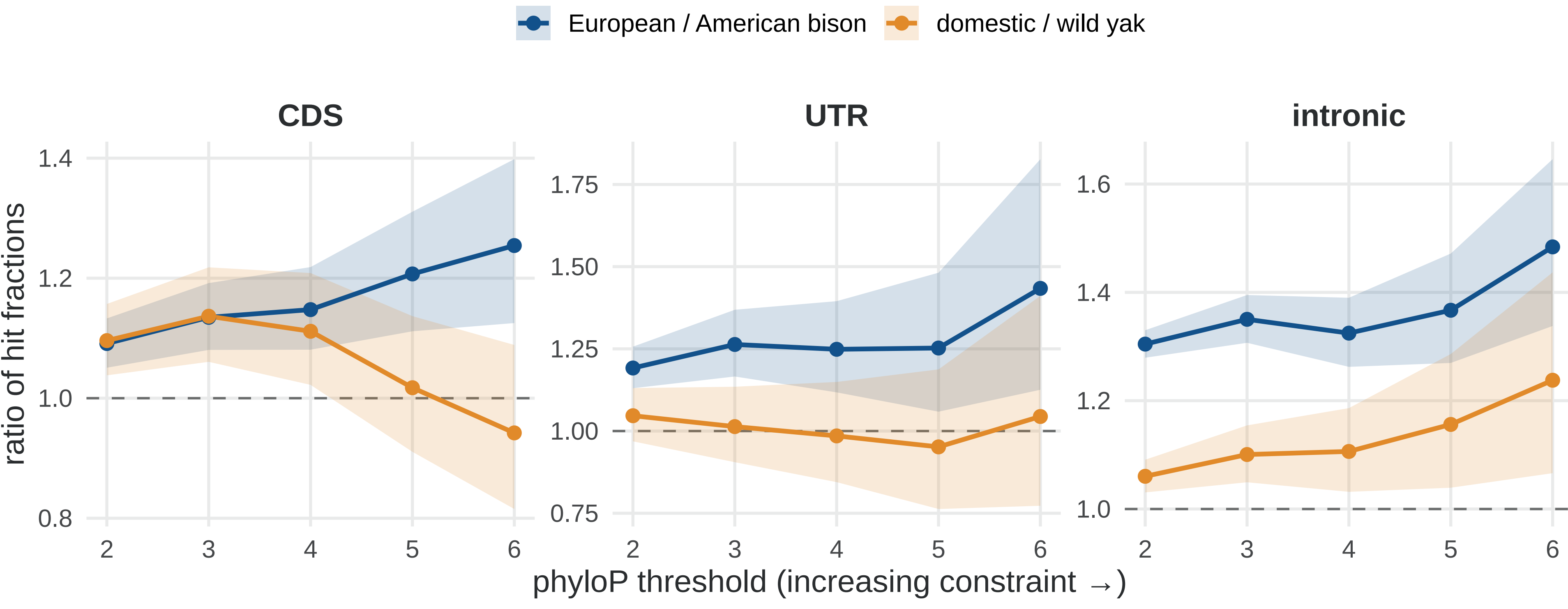
Scoring

- phyloP: a score measuring the rate of evolution relative to neutral sites
- Calculated for human genome based on 241-way mammalian alignment



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European bison changes in constrained regions



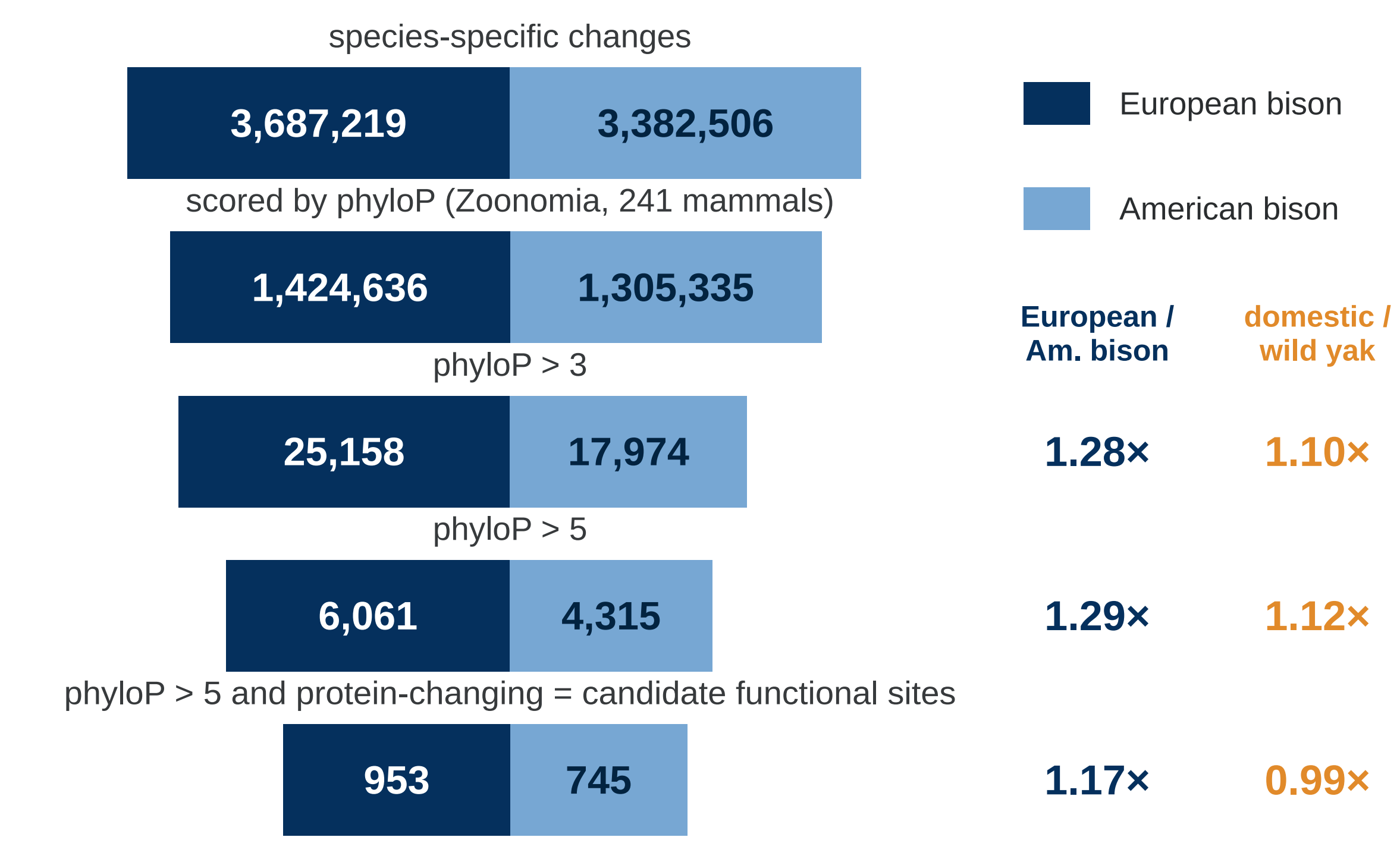
region	all changes		at phyloP > 3		ratio at phyloP > 3
	European bison <i>n_w</i>	American bison <i>n_b</i>	European bison <i>k_w</i>	American bison <i>k_b</i>	
CDS	21 242	18 669	3 153	2 442	1.13×
UTR	32 314	29 689	1 364	992	1.26×
intronic	617 919	565 039	8 808	5 965	1.35×

ratio = $\frac{k_w/n_w}{k_b/n_b}$

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Number of functional sites

- 953 candidate sites in 845 genes, 762 of them with exactly one.

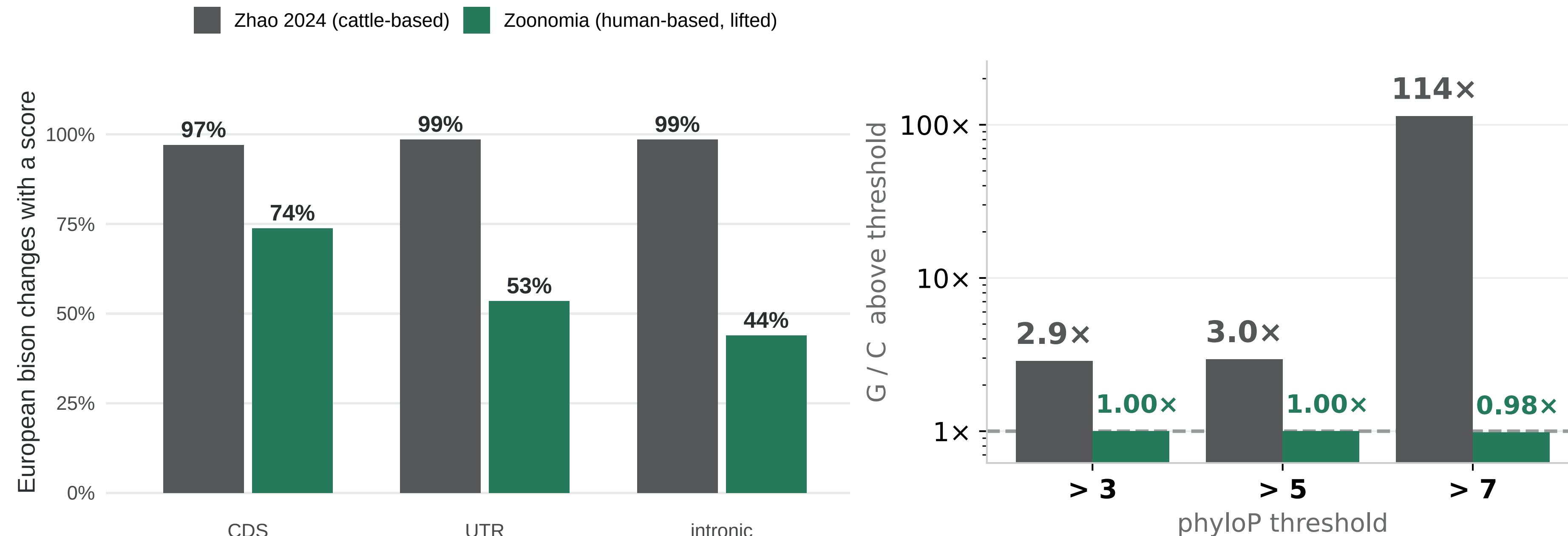


European / Am. bison	domestic / wild yak
1.28×	1.10×
1.29×	1.12×
1.17×	0.99×

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Can cattle-based scores improve predictions?

- New phyloP calculated based on the cattle genome covers 97-99% bison genome
- We found a reference-base bias: G sites pass high thresholds ~100× more often than C



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Conclusions

- European bison has changed more constrained sites than American bison, in every region and at every threshold
- Results suggest European bison has accumulated more changes at functionally important sites than American bison, especially in non-coding and highly constrained regions

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Future direction

- Improve cattle-based phyloP: fix the reference-base bias for full-coverage scoring
- Score candidates with genomic sequence models (AlphaGenome, Evo 2, GPN-Star), which reach the non-coding sites where the signal is the strongest
- Place candidate sites in gene function context: which functions are affected



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